

Research Article

Dynamic Contagion of Systemic Risks on Global Main Equity Markets Based on Granger Causality Networks

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Received 1 May 2018; Revised 15 July 2018; Accepted 19 July 2018; Published 7 August 2018

Academic Editor: Ricardo López-Ruiz

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A total of 156 Granger causal networks of stock markets are constructed by using the Granger causality test and time series sliding window based on stock index data of 34 major stock markets in the world from 2004 to 2017. The topological structures and evolution characteristics of the Granger causal networks are analyzed from the perspective of complex network theory. Empirical results demonstrate that the network topology has a significant difference during the global financial crisis and other periods. The causal relationships among different global stock markets exhibit a jump growth when each major crisis occurs. The contagion path is also short. A causal relationship between any two stock markets can usually be established with one stock market on average, not by using more than five stock markets. For risk contagion, the American stock markets exerted the largest influence in 12 years, followed by the European stock markets. Stock markets with high intermediate contagion ability play an important role in systemic risk contagion. Despite the crucial markets in Europe and America (e.g., USA, Brazil, and Mexico), stock markets with weak network correlation and strong media ability (e.g., the markets of Japan, Korea, Australia, and New Zealand) play a critical role in risk contagion.

1. Introduction

The American subprime mortgage crisis reignited the intense concern of economists regarding financial systemic risk. The stock market is the most important component in the entire financial system. Risks of a single stock market can spread to other correlated stock markets and even to the whole financial system, thus generating systemic risks. The strong risk contagion of stock markets can cause immense damage to the whole financial system, which may induce a financial crisis. Given the bankruptcy of subprime mortgage institutions and the forced closure of investment funds, the subprime crisis occurred in 2007, which then triggered violent fluctuations of the American stock markets. This subprime crisis swept global major stock markets in the European Union and Japan, which eventually led to the global financial crisis. Therefore, governments worldwide must strengthen supervision over stock markets, maintain

market stability, and prevent systemic risk during economic development.

Scholars in the 2008 conference on the “New Progresses on Research of Financial Systemic Risks” in London generally believed that research on financial systemic risks emphasizes the effects of the financial asset price fluctuation possessed by participants in financial market on the whole system. During a crisis, the interaction of such price fluctuation is further intensified by the significant convergence, correlation, and systemic risk contagion in the financial market. Given this background, understanding the complicated correlations among global major financial markets, propagation paths, and key nodes for propagation from the perspective of stock price is imperative to prevent systematic risk contagion. A scientific alarm system and monitoring mechanism are then established.

Systemic risk studies primarily employ three important measurement models: conditional value-at-risk (CoVaR) [1],

a method proposed by Adrian and Brunnermeier; the systemic expected shortfall (SES) [2], an approach suggested by Acharya et al.; and the distressed insurance premium (DIP) [3], a technique proposed by Huang, Zhou, and Zhu. These models measure systemic risks based on data at the crisis outbreak. During and after the occurrence of a financial crisis, financial institutions or financial markets show considerable high correlation. However, before the occurrence of a financial crisis, such a correlation seems hardly significant in the measurement of systemic risk. Research on systemic risk in the post-crisis era focuses on the correlation and comovement among financial institutions before and during the crisis.

Complex network [4–6] has been a popular research topic in recent years and has been widely applied in finance. Constructing an economic and financial complex network based on the financial time series (generally stock return series) can systematically and intuitively express mutual dependence among different financial institutions [7–10]. Unconditional direct relationships between all financial institutions and markets can be disclosed by the complex network, especially the relationships among financial institutions that have not suffered similar losses in the crisis. Eryigit et al. [11] established a complex network of 143 stock indexes of 59 countries by calculating the correlation coefficient among different stock yields by using the minimum spanning tree and the plane maximum filtering graph methods. They found that North American and European markets have the closest relationships, whereas the correlations among countries in Eastern Asian countries as well as between Eastern Asian and Western markets are weaker than other markets. Kenett et al. [12] established a network of well-capitalized 300 stocks in the New York Stock Exchange based on the partial autocorrelation among stock yields. Compared with the correlation coefficient, the partial correlation coefficient can measure the effects of one stock on the correlation coefficient between two other stocks. Such a correlation also has high application values for studying systemic risk contagion. Giuseppe et al. [13] analyzed 49 industries in the American equity markets from 1969 to 2011 and rendered mutation points of the average correlation coefficients of different industries corresponding to the financial crisis points, thus confirming the close correlation between risk contagion and stock correlation. Chunxia Y [14] conducted an in-depth analysis of 789 stocks in the global major stock markets based on the correlation price of stock prices. The systemic risk problem of stock markets in different stages is also studied. The empirical study reveals that a financial crisis may not change the correlation of stocks, but it will increase and then decrease the correlation coefficient.

There were also other good approaches for comovements modeling in finance, such as the technique of hierarchical clustering. Lahmiri S [15] dealt with the problem of Casablanca Stock Exchange (CSE) topology modeling as a complex network using hierarchical clustering linkage technique. The general structure of the CSE topology has considerably changed in 2009(variable regime), 2010 (increasing regime), and 2011 (decreasing regime). Lahmiri S [16] examined short and long-term dynamics in linkages

between global major markets during and after financial crisis based on the wavelet presentation of clustering analysis. The empirical results show strong evidence of the instability of the financial system aftermath of the global financial crisis

The approaches in the aforementioned literatures are mainly undirected correlation modeling. However, measuring not only the degree of connectedness between financial institutions but also the directionality of such relationships is important to investigate the dynamic propagation of shocks to the system. Therefore, constructing a directed complex network is necessary to recognize the source of systemic risk and the important nodes in the contagion. However, only a few studies related to the subject are reported. Billio M et al. [17] developed a directed causal network by using the yield data of hedge funds, banks, security traders, and insurance companies in the American financial market. The important role of banks in systemic risk contagion is also uncovered. Mensah J O et al. [18] established the Granger causal network of Asian banks through the CoVaR method and found that the correlations of Asian banks generally increase. The universal systemic risk is higher than those in other emerging markets. Lahmiri S [19] investigated cointegration and causal linkages among five different fertilizer markets during low and high market regimes. Fertilizer markets are closely linked to each other during low and high regimes. Jiang M [20] proposed an algorithm to transfer this evolution process to a complex network. Causality patterns are considered as nodes and the succeeding sequence relations between patterns as edges. The results show that a few types of causality patterns play a major role in the process of the transition and that international crude oil market is statistically and significantly not random. Li L [21] built a co-loan network to research the topological structures and corresponding evolvement characteristics of the Chinese banking system from 2008 to 2016. The co-loan network always displays a core-periphery structure.

Based on the preceding studies, an inference stating that a directed network better reflects systemic risk contagion than its undirected counterpart can be rendered. However, existing literature minimally addresses the directed and dynamic systemic risk contagion among global major stock markets. To this end, we propose using Granger causality measure of connectedness to construct the directed network. We can find Granger causality among price changes of financial assets in the presence of value-at-risk constraints or other market frictions such as transaction costs, borrowing constraints, costs of gathering and processing information, and institutional restrictions on short sales. The degree of Granger causality in asset returns can be viewed as a proxy for return-spillover effects among market participants. As this effect is amplified, connection and integration among financial institutions are tight, heightening the severity of systemic events [17]. Moreover, a Granger causality measure of connectedness can capture the lagged propagation of return spillovers in the financial system. Therefore, Granger causality network can best describe the systemic risk contagion among global major stock markets.

This study collected the stock indexes of 34 global major stock markets from 2004 to 2017. The source of systemic risk

contagion and important nodes according to the complicated relationships of yield spillover and systemic risk contagion among research markets in a directed causal network were also identified. Moreover, the evolution characteristics of the stock market network are explored.

The remainder of this paper is organized as follows. Section 2 introduces the research methodology and data source in the empirical analysis and presents the preliminary data processing. Section 3 analyzes and discusses dynamic risk contagion among global major stock markets according to the topological features of the 156 Granger causal networks constructed. Section 4 concludes.

2. Data and Research Methods

2.1. Data Selection and Processing. In this study, index data of 34 major stock markets in Asia, America, Europe, and Oceania from May 3, 2004, to June 30, 2017, were selected. Complete stock market index data are shown in the Appendix. To discuss dynamic contagion of systemic risk among different stock markets, the actual model used sliding window (length: three months and one month for each sliding) to select 156 subsamples and 156 constructed Granger causality network groups. For example, data in May, June, and July 2004 were selected as one subsample to discuss causal relations and systemic risk contagion among different stock markets in July 2004. The rest could be conducted in the same manner.

Stationary test of data is necessary before Granger causality test. Unit-root test is generally applied. Certain financial time series, such as volatility and return series, are stationary in most cases. Therefore, logarithmic returns of closing price were selected as research data. The formula is as follows:

$$r_i(t) = \ln P_i(t) - \ln P_i(t-1), \quad (1)$$

where $P_i(t)$ is the closing price of index i on t .

Before the analysis, return series of all 34 stock market indexes were examined by unit-root test [22]. The results show that each return series possessed no unit-root. All return series were stationary.

2.2. Granger Causality Test. In this study, mutual overflow relations among global major stock markets were recognized by Granger causality test [23, 24]. Time series j is said to “Granger cause” time series i if past values of j contain information that helps predict i above and beyond the information contained in the past value of i alone [17]. In one binary p -order VAR model,

$$\begin{aligned} \begin{pmatrix} y_t \\ x_t \end{pmatrix} &= \begin{pmatrix} \phi_{10} \\ \phi_{20} \end{pmatrix} + \begin{pmatrix} \phi_{11}^{(1)} & \phi_{12}^{(1)} \\ \phi_{21}^{(1)} & \phi_{22}^{(1)} \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} \\ &+ \begin{pmatrix} \phi_{11}^{(2)} & \phi_{12}^{(2)} \\ \phi_{21}^{(2)} & \phi_{22}^{(2)} \end{pmatrix} \begin{pmatrix} y_{t-2} \\ x_{t-2} \end{pmatrix} + \cdots \\ &+ \begin{pmatrix} \phi_{11}^{(p)} & \phi_{12}^{(p)} \\ \phi_{21}^{(p)} & \phi_{22}^{(p)} \end{pmatrix} \begin{pmatrix} y_{t-p} \\ x_{t-p} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} \end{aligned} \quad (2)$$

Variable x is not the Granger cause of y if and only if all coefficients $\phi_{12}^{(q)}$, ($q = 1, 2, \dots, p$) in the coefficient matrix are zero. This condition indicates that x cannot cause changes in y . Granger causality test examines whether lagged variable of one variable can be introduced into other variable equations. If variable x can help interpret variable y , then variable x is the Granger cause of variable y . F-test is a direct method to judge Granger cause:

$$\begin{aligned} H_0 : \phi_{12}^{(q)} &= 0, \\ q &= 1, 2, \dots, p \quad (3) \end{aligned}$$

$$H_1 : \text{at least there exist a } q \text{ such that } \phi_{12}^{(q)} \neq 0$$

The statistics are as follows:

$$S_1 = \frac{(RSS_0 - RSS_1)/p}{RSS_1/(T - 2p - 1)} \sim F(p, T - 2p - 1) \quad (4)$$

They conform to the F distribution. If S_1 is higher than the critical value of F, then the null hypothesis is rejected. Variable x is the Granger cause of variable y . Otherwise, the null hypothesis is accepted. RSS_1 is the residual sum of squares in the y equation (2): $RSS_1 = \sum_{t=1}^T \varepsilon_{1t}^2$, and RSS_0 is the residual sum of squares in the y equation when $\phi_{12}^{(q)} = 0$.

Granger causality test results are closely related to the number of lag orders. Therefore, the appropriate p in VAR model should be selected appropriately. On the one hand, p is expected to be sufficiently large to reflect dynamic features of the constructed model completely. On the other hand, large p bring several parameters for estimation, thereby reducing degrees of freedom (DOFs) of the model. Thus, comprehensive considerations to the quantity of lag items and DOFs are necessary when selecting the number of lag orders. In actual studies, Akaike Information Criterion (AIC) and Schwarz Criterion (SC) are common methods. The calculation formula is as follows:

$$\begin{aligned} AIC &= \frac{-2l}{T} + \frac{2n}{T} \\ SC &= \frac{-2l}{T} + n \frac{\ln T}{T} \end{aligned} \quad (5)$$

where n is the total number of estimated parameters, k is the number of endogenous variables, T is the sample length, d is the number of exogenous variables, and p is the number of lag orders. The logarithmic likelihood value l can be calculated by hypothesizing that the multivariate normal distribution is obeyed, as follows:

$$l = -\frac{Tk}{2} (1 + \ln 2\pi) - \frac{T}{2} \ln \left(\det \left(\frac{1}{T-m} \sum_t \varepsilon_t \varepsilon_t' \right) \right) \quad (6)$$

Low values of AIC and SC are ideal [24].

2.3. Granger Causality Network. The causality network at t $G_t = (V, E_t)$ was established through Granger causality test. Among them, the point set $V \subset N$ is listed companioned

(number of points is 34 at this moment), and the side set $E_t \subset V \times V$ contains all sides between any two points $i, j \in V$. If i is the Granger cause of j , then one directed side from i to j ($i \rightarrow j$) exists. Otherwise, no side exists. The following causality index function is defined:

$$C_{(i \rightarrow j)} = \begin{cases} 1, & \text{if } i \text{ is Granger Causality of } j \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

3. Empirical Results

The reasonable number of lag order was determined by AIC and SC in (5). The VAR model was established to test F statistics in (4), thereby obtaining causalities of daily return rates of 34 stock markets. The stock market indexes were used as network nodes. The Granger causality network groups (156 in total) that represent monthly causality from July 2004 to June 2017 were established in accordance with (7) by using 156 subsample data. Therefore, studying complicated causal relations of different stock markets is equal to studying causality network characteristics of stock market, including time-varying and important node characteristics.

Based on the established 156 Granger causality networks that change with time, important network nodes, and the differences of network topology before, during, and after financial crisis were analyzed by calculating out-degree and in-degree of network nodes. Other topological properties of each network were also examined.

3.1. Out-Degree and In-Degree of Granger Causality Networks. Given that Granger causality networks are directed networks, the following out-degree and in-degree of causality networks are defined:

$$\text{out-degree: } k^{\text{out}} = (j \rightarrow V) = \sum_{i \neq j, i \in V} (j \rightarrow i) \quad (8)$$

$$\text{in-degree: } k^{\text{in}} = (V \rightarrow j) = \sum_{i \neq j, i \in V} (i \rightarrow j) \quad (9)$$

where V is the point set in the entire network.

The definition shows that the out-degree and in-degree of nodes in causality networks can reflect the influence of different stock markets in the network. If one node possesses high out-degree, then it possesses more one-way causations than other nodes and the stock market is influential. If one node possesses high in-degree, then other nodes possess more one-way causations than this node and the stock market can be easily influenced by other markets.

Figure 1 shows the color code representations of the time evolution of out-degrees. Out-degrees of the American stock market were kept at a high level in most time, whereas out-degrees of the Asian and Oceania stock markets were low in most time. From the data statistics in Table 1, the American stock market was found with the highest average out-degrees and standard deviation (SD) in the past 14 years. Specifically, the New York Stock Exchange showed the maximum out-degree (17.11 in average). In terms of skewness and kurtosis of out-degrees, some values of most American and European

markets were close to 0 and approached to normal distribution with time. Monthly out-degree values are distributed at the two sides of the average symmetrically. Time-related distribution of out-degree values of the Asian and Oceania stock markets presented “peaks” and “right avertence.” This condition indicated that out-degrees concentrated at peak values and that such values were smaller than average values. The minimum out-degree was contributed by Shanghai Securities Composite Index in China. From the given statistical features, the American stock market was found to be the most influential in global stock markets, whereas the Asian and Oceania stock markets were the least influential. Out-degrees of the Asian and Oceania stock markets were lower than the average level in most time.

With regard to time-related evolution, the American stock market achieved the highest average out-degree during several periods of financial crisis, followed by the European, Asian, and Oceania stock markets successively. This finding implied the dominant role of the American stock market in the global stock market. The American stock market was the most influential, followed by the European stock market. For example, high out-degrees of Dow Jones Industrial Average (DJIA) in America mainly concentrated in three stages: US stock crash from May 2006 to August 2006, American subprime mortgage crisis and the global financial crisis it induced from 2007 to 2009, as well as the global stock crash from September 2015 to February 2016. The maximum out-degree occurred in the British and Spanish markets in Europe in April 2010 (Greek debt crisis) and in August 2011 (European debt crisis). Both markets replaced the American stock market to take the dominant role in the global market. Shanghai Securities Composite Index and Shenzhen Stock Index in Asia showed low out-degrees in most time, and they only showed high out-degrees during stock crashes in May 2006, March 2008, and August 2015.

Similarly, Table 1 reveals that the Asian (excluding China) and Oceania stock markets achieved the maximum in-degree mean and SD. Skewness of stock markets in most countries was smaller than 0. Thus, the time-related distribution of in-degrees showed “left avertence,” indicating that these stock markets were influenced significantly by other stock markets. The in-degree values of these markets were lower than the average level in most time.

The out-degrees and in-degrees of Chinese stock markets (Shanghai Securities Composite Index and Shenzhen Stock Index) were found to be low after comprehensive considerations. They were less influential in the global market and slightly influenced by other markets. This finding was related to China's stock market policies. The Chinese stock market remained separated from other world markets. Thus, it failed to attract tremendous external hot money. Moreover, China's regulations on foreign investors owning Chinese stocks shut the door upon many large international companies of asset management. However, out-degrees of China's stock markets for time-related variations increased sharply during stock crash. Moreover, the out-degrees and in-degrees of China's stock markets decreased to certain extents since 2015 (Figure 2). With the opening of Shanghai–Hong Kong and Shenzhen–Hong Kong Stock Connect as well as

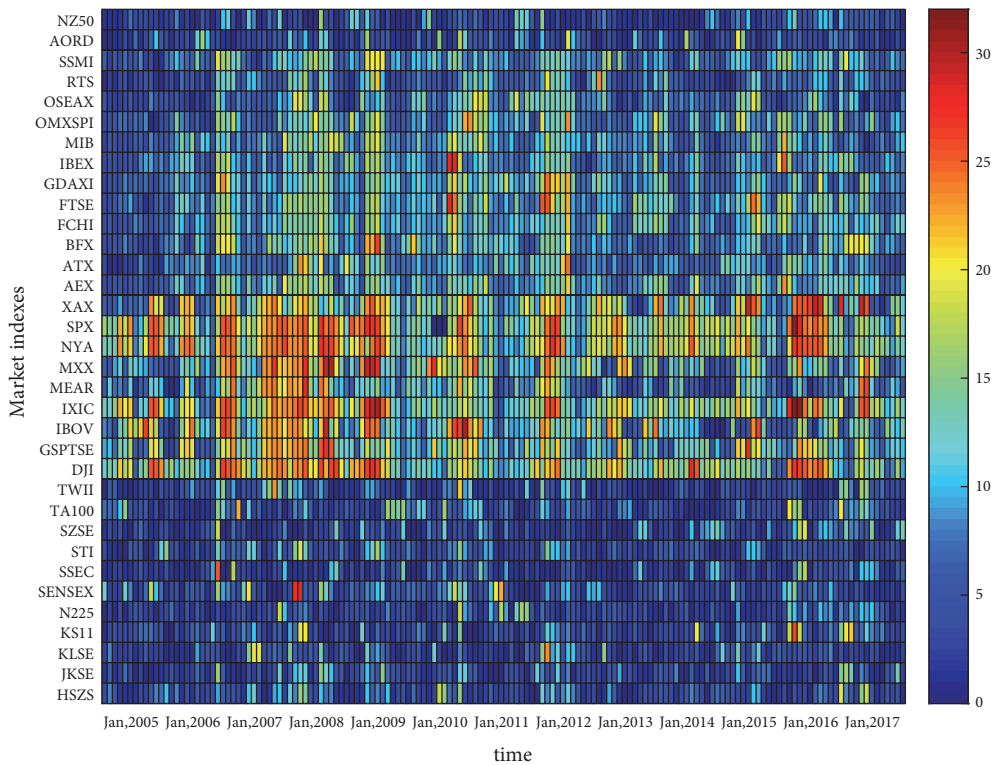


FIGURE 1: Color code representation of the time evolution of out-degrees.

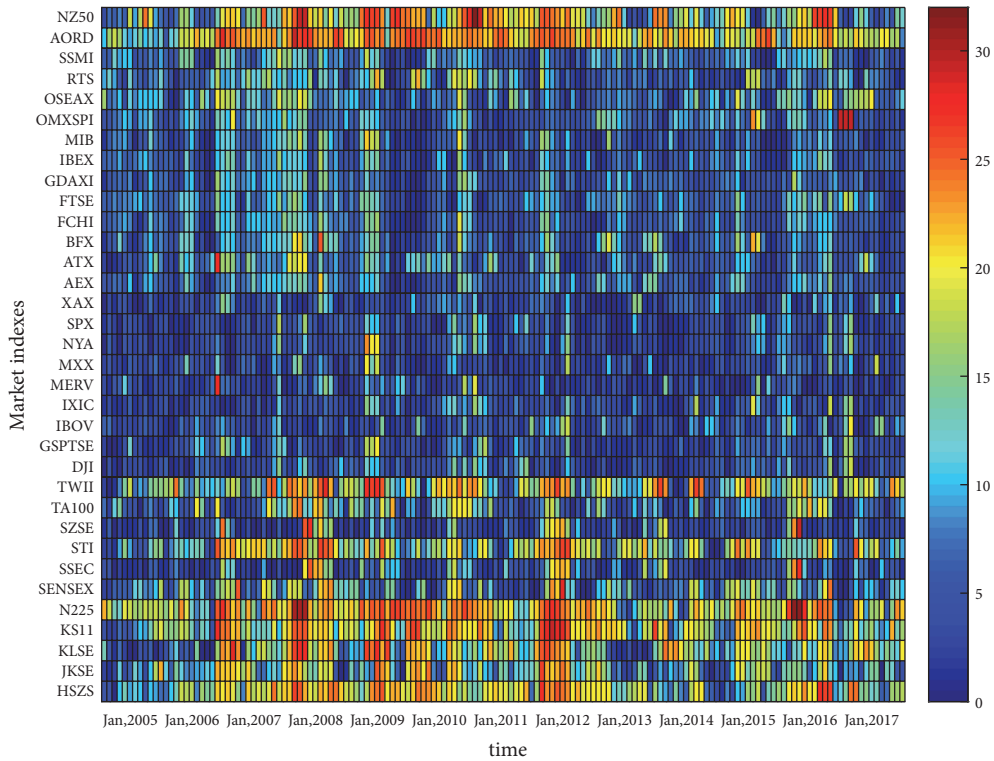


FIGURE 2: Color code representation of the time evolution of in-degrees.

Area	Code	Market	Out-degree				In-degree			
			Mean	SD	Skew.	Kurt.	Mean	SD	Skew.	Kurt.
Asia	HSZS	Hong Kong	4.44	3.82	1.14	0.95	16.43	6.03	-0.67	-0.34
	JKSE	Indonesia	3.86	3.52	1.28	1.30	12.41	6.69	0.02	-1.00
	KLSE	Malaysia	4.14	4.28	1.85	3.80	13.39	7.80	-0.12	-1.05
	KSII	Korea	4.51	4.44	2.04	5.11	17.05	6.34	-0.48	-0.49
	N225	Japan	3.66	3.67	2.01	4.42	19.55	5.65	-0.83	0.98
	SENSEX	India	5.81	5.25	1.75	3.81	9.59	6.45	0.47	-0.64
	SSEC	China	3.26	4.14	2.21	6.18	5.75	6.49	1.43	1.12
	STI	Singapore	4.70	3.93	1.16	1.06	14.65	6.63	-0.36	-0.69
	SZSE	China	3.54	3.93	1.64	2.74	5.26	6.51	1.76	2.41
	TAI100	Israel	4.87	4.49	1.35	1.54	8.74	6.34	0.55	-0.72
America	TWII	Taiwan	4.06	4.24	1.77	3.76	15.04	7.04	-0.25	-0.82
	DJI	America	16.64	5.77	0.03	-0.89	4.44	3.58	1.05	0.70
	GSPTSE	Canada	13.31	6.04	0.25	-0.76	4.49	4.19	1.46	1.96
	IBOV	Brazil	13.21	7.42	0.24	-0.97	4.64	3.72	1.22	1.35
	IXIC	America	16.42	6.09	0.24	-0.59	4.37	3.81	1.35	1.90
	MERV	Argentina	10.04	6.36	0.56	-0.30	3.94	4.29	2.16	6.52
	MXMX	Mexico	13.37	7.06	0.42	-0.63	3.98	3.82	1.73	3.51
	NYA	America	17.46	5.78	-0.18	-1.11	4.21	4.06	2.08	5.56
	SPX	America	17.30	6.14	-0.38	-0.17	4.12	3.66	1.32	1.68
	XAX	America	14.17	6.80	0.15	-1.04	4.06	3.64	1.16	0.96
Europe	AEX	Holland	9.89	4.30	0.40	-0.42	6.76	4.00	0.40	-0.13
	ATX	Austria	8.46	4.49	0.87	1.46	7.29	4.99	0.85	0.88
	BFX	Belgium	9.38	4.93	0.82	0.49	7.46	5.06	0.85	0.69
	FCHI	France	10.08	4.01	0.40	-0.37	6.51	4.16	0.48	-0.38
	FTSE	Britain	10.69	5.04	0.90	1.01	7.06	3.87	0.14	-0.65
	GDAXI	Germany	9.95	4.49	0.81	0.67	6.21	3.91	0.62	-0.11
	IBEX	Spain	9.24	4.64	1.34	2.96	6.11	4.05	0.42	-0.51
	MIB	Italy	9.42	4.29	0.55	-0.33	5.79	4.83	0.95	0.41
	OMXSPI	Sweden	9.30	4.93	0.63	-0.13	7.24	4.28	0.72	0.86
	OSEAX	Norway	7.76	4.92	0.60	-0.43	8.02	4.99	0.38	-0.36
Oceania	RTS	Russia	6.51	4.59	0.95	0.77	7.89	5.36	0.55	-0.36
	SSMI	Switzerland	8.84	4.26	0.46	-0.01	7.51	4.26	0.17	-0.51
	AORD	Australia	4.49	4.31	1.58	2.15	20.24	4.69	-1.01	0.58
NZ50	New Zealand	3.77	3.98	1.38	1.28	16.38	7.25	-0.29	-0.79	

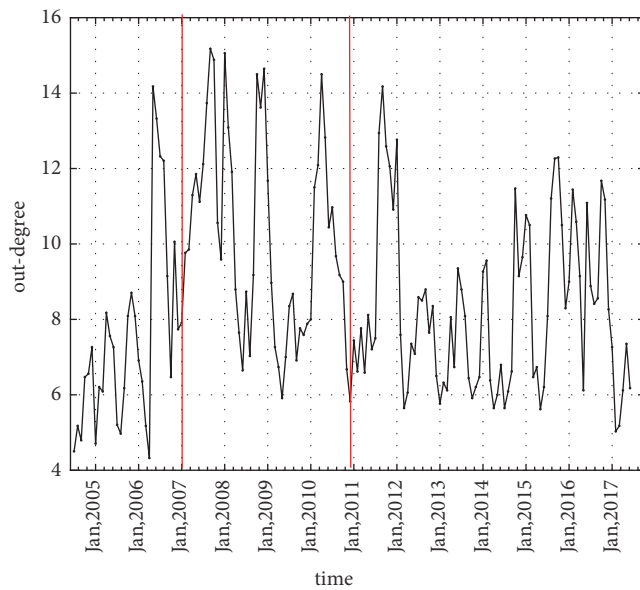


FIGURE 3: Dynamic evolutions of the monthly average out-degree of Granger causality networks.

enlisting RMB into International Monetary Fund's special drawing rights, China's capital market was further opened. This phenomenon might enhance the influence of China's capital market accordingly and might increase its sensitivity to the global capital market. Therefore, preventing systemic risk is important for China.

3.2. Difference of the Network Topology before, during, and after Financial Crisis. Figure 3 shows the dynamic evolutions of the monthly average out-degree of 34 nodes in the Granger causality networks. To examine the difference of the network topology before, during, and after financial crisis, we consider the most powerful global financial crisis of recent years. The whole time series is divided into three parts. The period corresponding to global financial crisis ranges from January 2007 to December 2010. The period before the global financial crisis ranges from July 2004 to December 2006. The period after the global financial crisis ranges from January 2011 to June 2017.

We examine the difference of average out-degree in three periods by analysis of variance (ANOVA), which is used to analyze the difference among group means in a sample. ANOVA is useful for testing three or more group means for statistical significance. Table 2 reveals that the networks have the largest mean, the largest maximum, and the largest minimum of monthly average out-degree during crisis. The mean of average out-degree went up after crisis, unlike the value before crisis. Table 3 presents the results of the difference comparisons among three different periods by ANOVA. We find that the difference is significant at the 0.05 level before and during the global financial crisis. The difference is also significant at the 0.05 level during and after the global financial crisis.

Apart from the global financial crisis, large and small financial crises occurred from July 2004 to June 2017. The

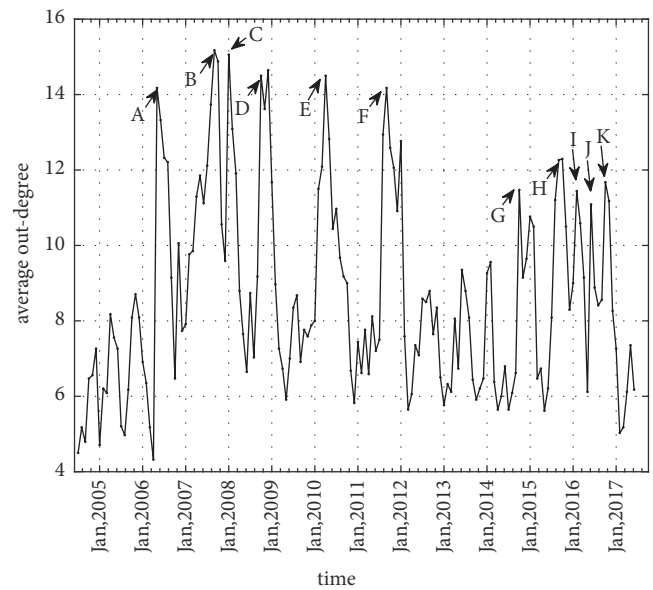


FIGURE 4: Dynamic evolutions of the monthly average out-degree label with points-in-time of great events occurred.

average out-degree changed quickly at the occurrence of each great event. Figure 4 shows that during the crash of global stock market (point A) in May 2006, the average out-degree from April 2006 to May 2006 soared and reached the peak since 2004. In January 2007, HSBC Holdings announced the business performance that the housing mortgage loan in North America had suffered great losses and the subprime crisis risk began. The average out-degree began to increase dramatically since then. As shown in Figure 4, the out-degree increased sharply at the comprehensive outburst of subprime crisis in August 2007 (point B), crash of stock market in Asian-Pacific region in January 2008 (point C), global financial crisis in October 2008 (point D), Greek debt crisis in April 2010 (point E), and European debt crisis in September 2011 (point F). The average out-degree increased to the local peak point. In October 2014, America exited from the quantitative easing policy (point G), which aroused international capital flow and intensified uncertainty of the international financial market. Out-degrees also reached the local peak at the crash of global stock market in September 2015 (point H). Black Swan events occurred frequently in 2016. In February 2016, Deutsche Bank released its annual report of 2015, which reported a loss of 6.8 billion EUR, causing violent market fluctuations. Deutsche Bank attracted high attention as the second Lehman Brothers. International Monetary Fund warned that among global banks that are systemically important, Deutsche Bank is the leading net contributor of systemic risk. In June, Britain voted to leave the European Union, which caused great fluctuation in the global stock market. American presidential election in October and November raised the risk aversion in the global financial market and capitals flooded into safe-haven assets massively, thereby resulting in the stock market crash. The tightness of global stock markets brought by Black Swan events increased. Figure 4 reveals that the average out-degree remained at a

TABLE 2: Summary statistics for the monthly average out-degree of three different periods.

Period	N	Mean	Std. Deviation	Std. Error	Minimum	Maximum
during crisis	48	10.0858	2.73009	0.39405	5.82	15.18
before crisis	30	7.4735	2.64365	0.48266	4.32	14.18
after crisis	78	8.2787	2.22409	0.25183	5.03	14.18
Total	156	8.6799	2.64571	0.21183	4.32	15.18

TABLE 3: Comparisons of multiple differences among three different periods by ANOVA.

(I) group	(J) group	Mean Difference (I-J)	Std. Error	Sig.
during crisis	before crisis	2.61225*	0.57500	0.000
	after crisis	1.80713*	0.45323	0.000
before crisis	during crisis	-2.61225*	0.57500	0.000
	after crisis	-0.80513	0.53076	0.131
after crisis	during crisis	-1.80713*	0.45323	0.000
	before crisis	0.80513	0.53076	0.131

*The mean difference is significant at the 0.05 level.

high level during the Black Swan events (points I, J, and K). However, central banks in the world “adopt all means” in 2016 to cope with the situation. Monetary policy reached the extreme loose state. Moreover, the soaring oil price brought a bullish market. The fluctuation was intensifying, but many stock markets achieved satisfying growth.

Causality networks of stock markets in four special periods are shown in Figure 5. Each causality network was divided into four parts through clustering. The upper left is the Oceania stock market, the upper right is the American stock market, the lower left is the Asian stock market, and the lower right is the European stock market. The minimum causalities among different stock markets were observed in April 2006 (Figure 5(a)). The four states exhibited average connections. In September 2007, connections of global stock markets reached the peak with the further outburst of subprime crisis (Figure 5(b)). During this period, nine stocks of the American stock market showed maximum influences, indicating that the subprime crisis of America influences the entire world. During the outburst of global financial crisis in October 2008, all stock markets were tied up closely, and the American stock market maintained strong influence. The European stock market joined in the systemic risk contagion (Figure 5(c)). During the European debt crisis in September 2011, the European and American stock markets strongly influenced the global economy (Figure 5(d)).

In conclusion, the network topology had a significant difference during the global financial crisis and other periods. Moreover, the network topology changed quickly at the every great event. Out-degrees reached local peak in most cases. Therefore, different stock markets tied up tightly during crises, thereby showing strong systemic risk contagion. Based on the network data, the American stock market took the dominant role during the American subprime crisis, indicating that this crisis might further spread and finally cause global financial crisis.

3.3. Average Shortest Path of Granger Causality Networks.

In Granger causality network, the number of sides on the shortest path between two nodes i and j is called the distance d_{ij} . The average path length of the entire network (L) is then defined as the average distance between any two nodes as follows:

$$L = \frac{1}{(1/2)N(N-1)} \sum_{i \neq j} d_{ij} \quad (10)$$

where N is the number of nodes in the network. The network diameter is the maximum L . The average path length is important network characteristics measure between all pairs of nodes in the network. The distance between nodes here refers to the minimum number of edges needed to connect two nodes. The average path length and diameter measure the transmission performance and efficiency of the networks. Therefore, the distance of the systemic risk contagion between any two major stock markets can help us understand the contagion path.

Table 4 shows the average shortest paths and diameters of the Granger causality network in several special periods in 140 months. The contagion path between different stock markets was the shortest during the American subprime mortgage crisis. The longest contagion path was observed in April 2006, during which the minimum average degree was found. The average shortest paths ranged between one and three in several special periods, and a causal relationship between any two stock markets could be established through one stock market on the average, not by using five stock markets at most. This finding revealed the close correlations among different stock markets and outstanding spillover effects. Therefore, the global market can be easily influenced when one market is in trouble, thereby breaking out systemic risk crisis.

3.4. Betweenness Centrality of the Granger Causality Network.

If a stock market stands in the shortest path between two

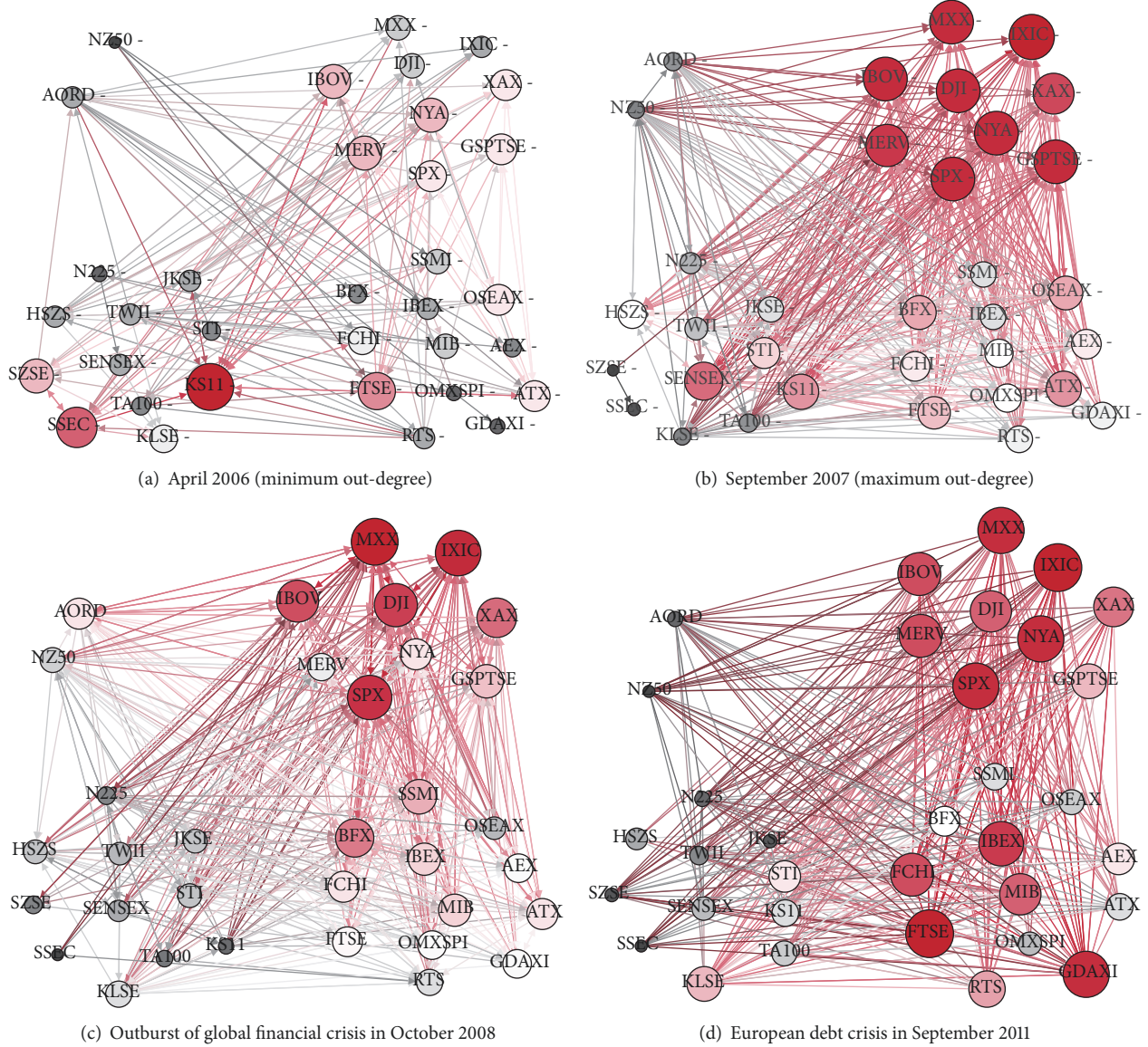


FIGURE 5: Causality networks of stock markets in four special periods.

markets, then it plays the role of transmission medium in the transmission process. The level of the transmission media effect can reflect the ability to control information in the transmission process. Only by transmitting through these media can certain stock markets transmit to others. Transmission media play an important role in the topological structure of complex networks [25].

Thus, the normalized betweenness centralities of nodes, which can denote the media ability of each stock market are evaluated. Betweenness centrality is an important network topological property, which depicts importance of nodes by the number of the shortest paths passing through one node. The normalized betweenness centrality BC_i of node i can be defined as follows: $BC_i = (\sum_{s \neq i \neq t} (n_{st}^i / g_{st})) / (N^2 - 3N + 2)$, where g_{st} is the number of the shortest paths from node s to node t , and n_{st}^i is the number of the shortest paths from

node s to node t , which passes through the node i . The higher the normalized betweenness centralities, the stronger the transmission media ability.

Betweenness centralities of each node in 156 causality networks were calculated. Figure 6 shows that the color code represents the value of normalized betweenness centrality of main stock market indexes. The main color is blue, which means that most stock market indexes presented small normalized betweenness centrality from July 2004 to June 2017. Only a few stock markets showed large betweenness centrality and were used as leading transmission media. For example, the values of the stock market indexes AORD, N225, and KSII are higher than the others in most months.

Figure 7(a) indicates that 29.41% of stock market indexes shouldered 73.91% of the transmission media ability in September 2007 during which the maximum network degree

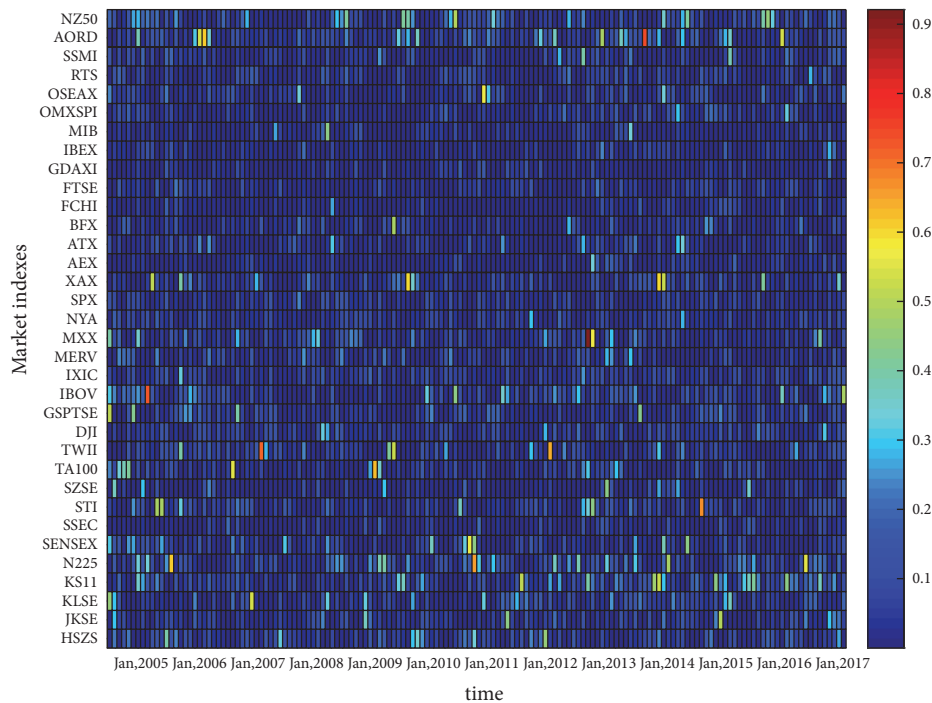


FIGURE 6: Color code representation of betweenness centrality of Granger causality networks.

TABLE 4: Average shortest paths and diameters of the Granger causality networks.

time	Average degree	Average shortest path	diameter
April 2006 (The minimum out-degree)	4.324	2.474	6
September 2007 (The maximum out-degree)	15.177	1.455	3
October 2008 (global financial crisis)	14.500	1.585	3
September 2011 (the European debt crisis)	14.177	1.641	3
September 2015 (the crash of global stock market)	11.765	1.765	4
Mean	8.723	1.827	4

was observed. The top three stock market indexes also undertook 30.09% media ability in the network. In April 2006 during which the minimum network degree was found, 26.47% of stock market indexes shouldered 70.76% of the transmission media ability (Figure 7(b)), and the top three stock market indexes undertook 33.10% of the media ability in the network. Similarly, 26.47% stock market indexes shouldered 73.91% of the transmission media ability (Figure 7(c)) during the European debt crisis in September, 2011. Moreover, the top three stock market indexes undertook 42.44% of the media ability in the network. Therefore, these stock markets with transmission media ability play an important role in information exchange and transmission, especially in risk contagion, around the world.

Figure 8 reveals the graphs of average betweenness centrality of each stock market and the corresponding average out-degree (W^{out}) in the past 14 years. Three stock indexes (zone B), including IBOV ($BC = 0.0847$, $W^{out} = 13.21$), showed strong transmission media ability (high betweenness centrality) and high contagion ability (high out-degree). These indexes absolutely played important roles in the entire

network. Moreover, a type of indexes (zone C), such as NYA ($BC = 0.0461$, $W^{out} = 17.4571$), presented high out-degree. Although these indexes exhibited poor transmission media ability, they possessed strong output capability of systemic risk. They could not be neglected. A type of stock market indexes (zone C) with low out-degree and high betweenness centrality, such as AORD ($BC = 0.1090$, $W^{out} = 4.4929$), presented strong transmission media ability. They could be easily neglected. These indexes were not the most influential ones in the global market, but they still played important roles in systemic risk contagion. Table 5 shows the specific numerical values.

4. Conclusions

The complicated causal correlations of global stock markets in 12 years are clearly described based on 156 Granger causal networks. Findings reveal that the network topology has a significant difference during the global financial crisis and other periods, and the causal relationships of global stock markets show a jump growth at the occurrence of crises.

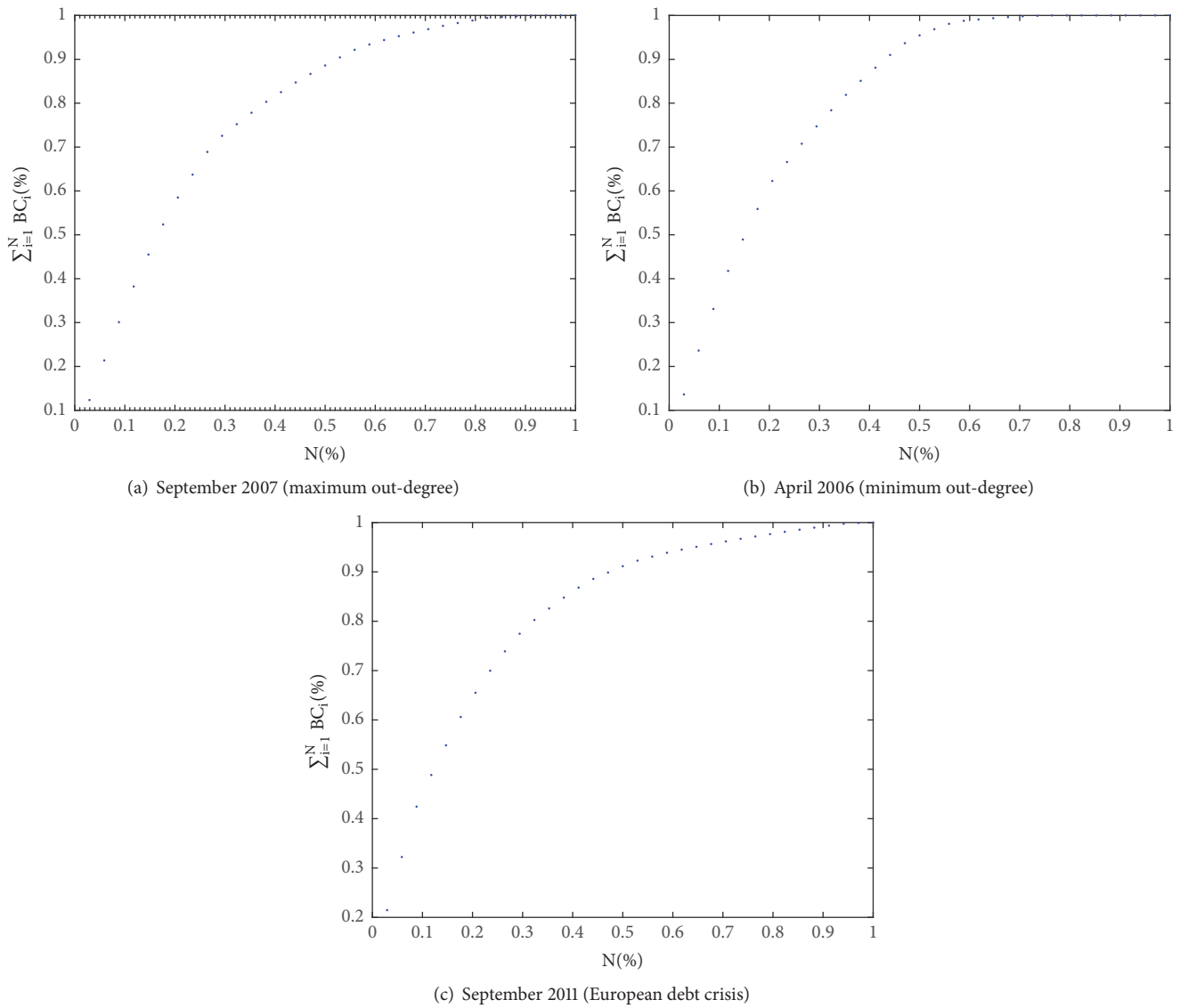


FIGURE 7: Distributions of normalized betweenness centrality of Granger causality networks.

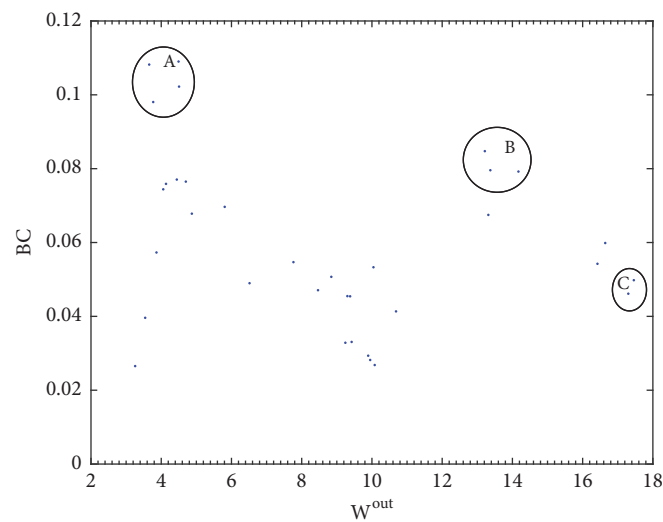


FIGURE 8: Distributions between out-degree and normalized betweenness centrality of nodes in descending order.

TABLE 5: Three types of market index.

type	Location on Figure 8	Market index	BC	W^{out}
higher out-degree higher media ability	B	IBOV	0.0847	13.2143
		MXX	0.0796	13.3714
		XAX	0.0792	14.1714
higher out-degree lower media ability	C	NYA	0.0461	17.4571
		SPX	0.0498	17.3000
		AORD	0.1090	4.4929
lower out-degree higher media ability	A	N225	0.1082	3.6571
		KSII	0.1022	4.5071
		NZ50	0.0980	3.7714

American stock markets have been occupying the dominant role in risk contagion, followed by European stock markets. Moreover, the path of the systemic risk contagion path is short. A causal relationship between any two stock markets can usually be established with one stock market on the average, but not by using more than five stock markets, thus showing the strongest systemic risk contagion. Moreover, key nodes in the systemic risk contagion of networks of global stock market and three types of important markets of systemic risk contagion are ascertained, especially the third type of markets that are easily overlooked (e.g., the markets of Japan, Korea, Australia, and New Zealand). Members of the third market type have low influence, but they possess strong intermediate conduction ability. These results have important implications for recognizing and preventing financial systemic risks. To prevent further risk contagion, these major stock markets are advised to establish effective alarm mechanisms that apprehend the “black swan” events as soon as possible. They should also pay great attention to other stock markets with strong network correlations. This research has significant limitations in terms of data availability and integrity. Only 34 stock markets with complete yield data in 12 years are selected. Exhaustive and real-time updated data are necessary to always interpret the changes and development of the global financial systemic risk. Except for yield rate, the mutual influences of other factors (e.g., mobility) among stock markets may be explored in future studies. A contrast analysis on these factors should be conducted, and the outcomes of systemic risk contagion in global stock markets should be described more comprehensively.

Appendix

Index data are as follows: (1) Asia: Hang Seng Index (HSZS), Indonesian JKSE Index (JKSE), Malaysian KLSE Index (KLSE), Korean KOSPI Index (KSII), Nikkei 225 Index (N225), BSE Sensex Index (SENSEX), Shanghai Composite Index (SSEC), Singapore Straits Times Index (STI), China Shenzhen Composite Index (SZSE), Israel TA100 Index (TA100), and the Taipei Weighted Index (TWII); (2) America: Dow Jones Industrial Average (DJI), S&P-TSX Composite Index of Toronto, Canada (GSPTSE), Brazil Bovespa

Index (IBOV), American NASDAQ Index (IXIC), Argentina MerVal Index (MERV), Mexico IPC Index (MXX), New York Stock Exchange Composite Index (NYA), Standard & Poor's Indices (SPX), and the American Stock Exchange Index (XAX); (3) Europe: Netherlands AEX Index (AEX), Austria ATX Index (ATX), Belgium BFX Index (BFX), France CAC40 Index (FCHI), London Financial Times 100 Index (FTSE), Germany Frankfurt DAX Index (GDAXI), Spain IBEX35 Index (IBEX), Italy FTSE MIB Index (MIB), Sweden OMX Index (OMXSPI), Norway OSEAX Index (OSEAX), Russia RTS Index (RTS), and the Switzerland SWI20 Index (SSMI); and (4) Oceania: Australian Ordinaries Index (AORD) and the New Zealand 50 Index (NZ50).

Data Availability

The data used to support the findings of the study are stock index yield data in the world's major stock markets. The data can be accessed from <http://www1.resset.cn:8080/product/> if you pay and get an account. The data used to support the findings of the study are included within the supplementary information file.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (no. 71171135) and Shanghai First Class Discipline Project (no. S1201YLXK).

Supplementary Materials

The supplementary material is an excel file which is the data used to support the findings of the study. (*Supplementary Materials*)

References

- [1] T. Adrian and M. K. Brunnermeier, “CoVaR,” *American Economic Review*, vol. 106, no. 7, pp. 1705–1741, 2016.

- [2] V. V. Acharya, L. H. Pedersen, T. Philippon, and M. Richardson, "Measuring Systemic Risk," *CEPR Discussion Paper*, vol. 29, no. 1002, pp. 85–119, 2010.
- [3] X. Huang, H. Zhou, and H. Zhu, "A framework for assessing the systemic risk of major financial institutions," *Journal of Banking & Finance*, vol. 33, no. 11, pp. 2036–2049, 2009.
- [4] D. J. Watts and S. H. Strogatz, "Collective dynamics of "small-world" networks," *Nature*, vol. 393, no. 6684, pp. 440–442, 1998.
- [5] A. Barabasi and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, 1999.
- [6] E. Jin M, M. Girvan, and E. Newman M, "Structure of growing social networks," *Physical Review E: Statistical Nonlinear & Soft Matter Physics*, vol. 64, no. 4Pt2, pp. 322–333, 2001.
- [7] W.-Q. Duan and H. E. Stanley, "Cross-correlation and the predictability of financial return series," *Physica A: Statistical Mechanics and its Applications*, vol. 390, no. 2, pp. 290–296, 2010.
- [8] Y.-W. Lai, "Measuring rank correlation coefficients between financial time series: a GARCH-copula based sequence alignment algorithm," *European Journal of Operational Research*, vol. 232, no. 2, pp. 375–382, 2014.
- [9] L. Seemann, J.-C. Hua, J. L. McCauley, and G. H. Gunaratne, "Ensemble vs. time averages in financial time series analysis," *Physica A: Statistical Mechanics and its Applications*, vol. 391, no. 23, pp. 6024–6032, 2012.
- [10] W. Shi, P. Shang, J. Wang, and A. Lin, "Multiscale multifractal detrended cross-correlation analysis of financial time series," *Physica A: Statistical Mechanics and its Applications*, vol. 403, pp. 35–44, 2014.
- [11] M. Eryigit and R. Eryigit, "Network structure of cross-correlations among the world market indices," *Physica A: Statistical Mechanics and its Applications*, vol. 388, no. 17, pp. 3551–3562, 2009.
- [12] D. Y. Kenett, M. Tumminello, A. Madi, G. Gur-Gershgoren, R. N. Mantegna, and E. Ben-Jacob, "Dominating clasp of the financial sector revealed by partial correlation analysis of the stock market," *PLoS ONE*, vol. 5, no. 12, Article ID e15032, 2010.
- [13] G. Buccheri, S. Marmi, and R. N. Mantegna, "Evolution of correlation structure of industrial indices of U.S. equity markets," *Physical Review E: Statistical, Nonlinear, and Soft Matter Physics*, vol. 88, no. 1, pp. 493–494, 2013.
- [14] C. Yang, X. Zhu, Q. Li, Y. Chen, and Q. Deng, "Research on the evolution of stock correlation based on maximal spanning trees," *Physica A: Statistical Mechanics and its Applications*, vol. 415, pp. 1–18, 2014.
- [15] S. Lahmiri, "Clustering of Casablanca stock market based on hurst exponent estimates," *Physica A: Statistical Mechanics and its Applications*, vol. 456, pp. 310–318, 2016.
- [16] S. Lahmiri, G. S. Uddin, and S. Bekiros, "Clustering of short and long-term co-movements in international financial and commodity markets in wavelet domain," *Physica A: Statistical Mechanics and its Applications*, vol. 486, pp. 947–955, 2017.
- [17] M. Billo, M. Getmansky, A. W. Lo, and L. Pelizzon, "Econometric measures of connectedness and systemic risk in the finance and insurance sectors," *Journal of Financial Economics*, vol. 104, no. 3, pp. 535–559, 2012.
- [18] J. O. Mensah and G. Premaratne, "Systemic interconnectedness among Asian Banks," *Japan and the World Economy*, vol. 41, pp. 17–33, 2017.
- [19] S. Lahmiri, "Cointegration and causal linkages in fertilizer markets across different regimes," *Physica A: Statistical Mechanics and its Applications*, vol. 471, pp. 181–189, 2017.
- [20] M. Jiang, X. Gao, H. An, H. Li, and B. Sun, "Reconstructing complex network for characterizing the time-varying causality evolution behavior of multivariate time series," *Scientific Reports*, vol. 7, no. 1, article 10486, 2017.
- [21] L. Li, Q. Ma, J. He, and X. Sui, "Co-Loan Network of Chinese Banking System Based on Listed Companies' Loan Data," *Discrete Dynamics in Nature and Society*, vol. 2018, no. 6, Article ID 9565896, 7 pages, 2018.
- [22] J. Geweke, R. Meese, and W. Dent, "Comparing alternative tests of causality in temporal systems: analytic results and experimental evidence," *Journal of Econometrics*, vol. 21, no. 2, pp. 161–194, 1983.
- [23] C. W. J. Granger, "Investigating causal relations by econometric models and cross-spectral methods," *Econometrica*, vol. 37, no. 3, pp. 424–238, 1969.
- [24] C. W. Granger, "Testing for causality: a personal viewpoint," *Journal of Economic Dynamics and Control*, vol. 2, no. 4, pp. 329–352, 1980.
- [25] X. Y. Gao, H. Z. An, W. Fang, X. Huang, H. Li, and W. Q. Zhong, "Characteristics of the transmission of autoregressive sub-patterns in financial time series," *Scientific Reports*, vol. 4, article 6290, 2014.